

PHLOTON

A public-good last-mile cold chain: active, solar-charged, recorded on every trip, and on the WHO PQS route.

THE DEVICE, IN THREE WORDS

Active. Off-grid. Connected.

A solar-charged, connected carrier that holds vaccines, blood and biologics at 2–8 °C where the grid ends. Fifty units in service in India; built for the world.

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2 L units in assembly, Bengaluru

Coverage stalled near 80%.

The last mile is where it stalls.

UNICEF delivered 3.2 billion doses to 103 countries in 2025, and 13.5 million children still received no vaccine at all. Gavi's 2026–2030 supply-chain strategy names cold-chain efficiency and equitable reach as priorities. The final kilometres still run on ice packs, freezers and paper.

3.2^B

vaccine doses UNICEF delivered to 103 countries in 2025 – all of them cross a last mile

UNICEF Annual Results, 2025

13.5^M

children received no vaccine at all in 2025

WHO / UNICEF, July 2026

1^B

people in lower-income countries served by health facilities without reliable power

WHO • World Bank • IRENA, 2023

WHAT A PUBLIC-GOOD DEVICE LOOKS LIKE

Active cooling without a freezer chain, charged by the sun, **no refrigerant at all**, every reading recorded. Built in India at \$700 today and \$500 at scale; on the WHO PQS E003 route so UN agencies and national programmes can buy it.



WHAT WE ASK A FUNDER TO BUY

Certification and evidence, not marketing: the PQS E003 submission, **100 units into two public health systems**, and a 30-versus-30 comparative field study against ice-pack carriers under an ethics-committee-approved protocol, with the data published.

Half of what the world makes never protects anyone.

By WHO's upper estimate, up to half of all vaccines are wasted every year; IQVIA puts the loss to temperature alone at 7–25%. About a billion people in lower-income countries depend on health facilities without reliable power. A cold chain that needs a grid and a freezer cannot follow the medicine to the patient.

50%

of vaccines wasted every year at WHO's upper estimate; up to a quarter lost to temperature alone

WHO, via UNEP 2020 • IQVIA 2025

\$35^B

lost by biopharma yearly to temperature failures

IQVIA Institute estimate, 2019

1^B

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WHO • World Bank • IRENA, 2023

HOW THE LAST MILE IS SOLVED TODAY

Ice-pack and phase-change-material (PCM) carriers, conditioned in a freezer upstream — or a mains-charged active carrier at about \$2,000.

WHY THAT FALLS SHORT

A passive carrier needs a working freezer, loses about half of what is left each time it is opened (our field rule of thumb), can freeze the payload, and records nothing. A mains-charged active carrier has to come back to a socket to recharge; the grid sets its range.

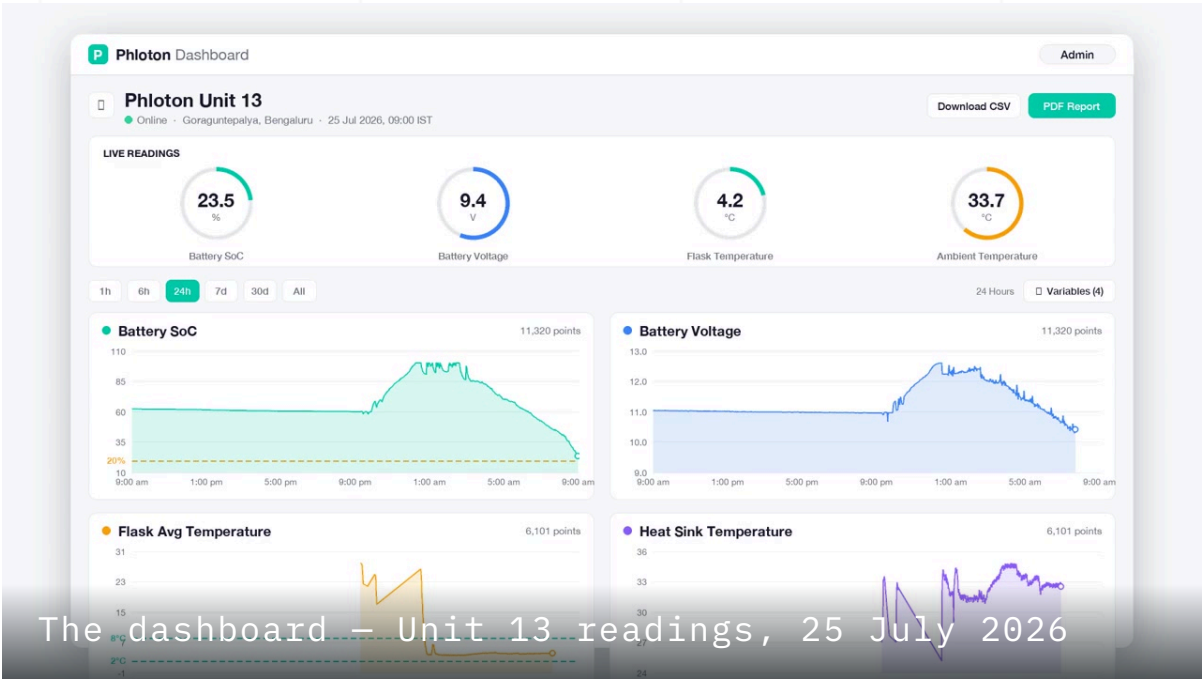
2–8 °C. Designed for sixteen hours at 43 °C. Charged by the sun.

A 3.75 kg, 2-litre device with a solid-state cooling engine: no ice, no freezer, no refrigerant. It holds 2–8 °C, charges from the sun, a vehicle or the wall, and reports every reading over 4G. From a warm start an empty chamber is inside 2–8 °C in fifteen to twenty minutes; the status light turns green when the chamber is in band.

AMBIENT • DESIGN	PAYLOAD • TARGET	HOLD • DESIGN	AUDIT TRAIL
43 °C	3 °C	16 h	Every trip

WHAT THE CUSTOMER SEES

Sixteen sensing points reported over 4G, with an offline buffer for the stretches without signal: each unit's readings and temperature trace, alerts when a reading drifts, the fleet on a map, and a compliance report at the end of every trip.



One engine. Every payload that must stay cold.

Every proven row below runs on the same solid-state cooling engine and controller. A new payload is a new customer, not a new engine; the larger sizes add a coolant loop.

PAYLOAD	HOLD BAND	WHERE IT STANDS
PROVEN IN THE FIELD		
Vaccines	2–8 °C	SELCO Foundation outreach clinics · PCM jackets for open-vial sessions
Blood bags	2–6 °C · never freezes	Raktamitra blood bank, Government Institute of Medical Sciences (GIMS), Noida — transported with no haemolysis (no red-cell damage)
Diagnostic samples	2–8 °C	Beckman Coulter — six months of 7–8 h runs, then a purchase
Specialty biologics	2–8 °C	Roche India, Kolkata — cold-chain-as-a-service, depot to hospital · starting
Veterinary & dairy	2–8 °C	Akshayakalpa · URMUL
NEXT — SAME ENGINE, NEW SET POINT OR FORM FACTOR		
Insulin & antivenom	2–8 °C	2.5 L for primary health centres
Reagents, enzymes, trial kits	programmable	5 L for labs and clinical-trial logistics
Organs & tissue	4–10 °C	10 L · exploratory, with a transplant partner

Active, off-grid, connected. Only one does all three.

The last mile needs three things: active cooling, power without a grid, and a live record. The table scores each option against all three.

	ICE-PACK CARRIER	PCM CARRIER	BLACKFROG EMVOLIO	PHLOTON
Price	~\$75 prequalified + a freezer	varies + a freezer	~\$2,000	\$700 → \$500 at scale
Holdover	set by the ice packs	fixed window, halves with each opening	12 h, verified at 43 °C	16 h at 43 °C · 24 h at 27 °C (design)
Weight · battery	—	—	5.5 kg · 200 Wh	3.75 kg · 390 Wh
Charging	needs a freezer	needs a freezer	mains	solar, vehicle or mains
Monitoring	none	none	live tracking	16 sensing points · 4G with offline buffer · per-trip compliance report
WHO PQS prequalification	listed (E004)	varies	listed (E003)	on the E003 route — Emvolio's category — funded by this round

WHO PQS is the prequalification UN agencies and immunisation programmes buy against: E003 covers powered cooling devices, E004 insulated containers
Patent filed · CDSCO (India's medical-device regulator) test licence · no refrigerant at all · five-year cost of ownership about a third below passive, on our model

\$700 device. \$60 a year. 5-year warranty.

Two revenue lines — the device, with a five-year warranty, and a service plan — and, for institutions, the whole thing as a contract.

LINE	PRICE	WHAT'S INCLUDED
Device • 2 L	\$700 today \$500 at scale	Device, charger and five-year warranty. Sold near cost at pilot volumes; at scale, \$500 against a build of about \$200 after enclosures are injection-moulded.
Service plan	\$60 per device per year	4G telemetry, alerts, per-trip compliance reports, maintenance. About 14% of revenue by year five.
Cold-chain-as-a-service	Priced per contract	We carry the payload and own the audit trail; they own the patient. First engagement starting with Roche India, Kolkata: depot to hospital.

\$1,000

Lifetime value per device over five years: \$700 device + 5 × \$60 service.

60%

Gross margin at scale — \$500 price against a ~\$200 build. Near cost today.

B2B

For three years — pharma, diagnostics chains, labs and healthcare institutions, reached through pharma distributors and hospital groups. Consumer comes later, by choice.

Fifty units in service.

All fifty bought.

Forty bought commercially, ten through grant programmes. Beckman Coulter ran diagnostic samples on 7–8 hour routes for six months, reported no excursions and no sample loss — then bought.

96%

of 1,883 monitored hold-hours in 2–8 °C – chamber telemetry from 17 connected units, January–August 2026

0

readings below 0 °C, fleet-wide – nothing freezes

0

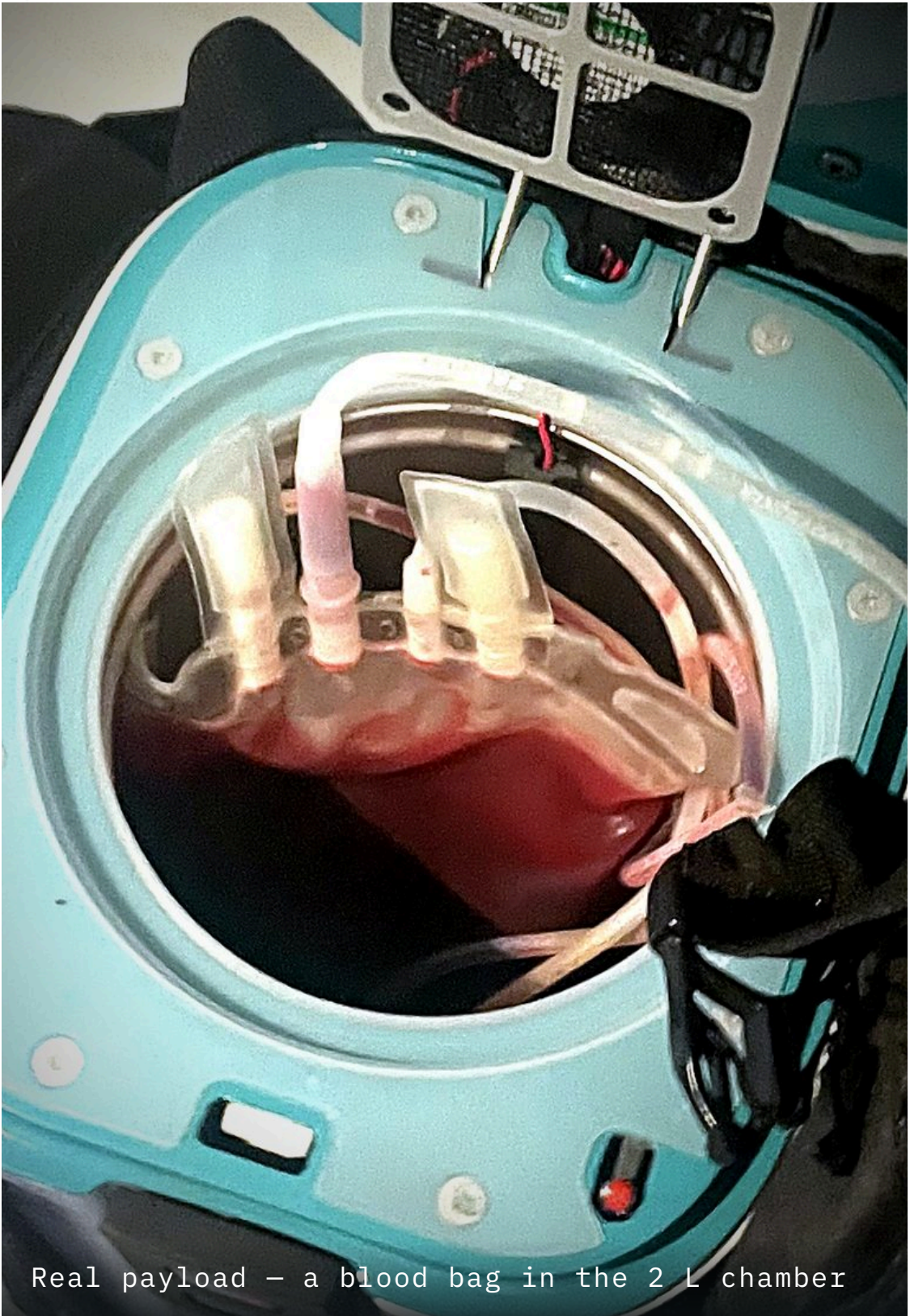
excursions reported in six months of 7–8 h sample runs at Beckman Coulter

IN HAND

CDSCO (India's medical-device regulator) test licence · patent filed · EMI/EMC testing at TÜV SÜD to IEC 61326-1 · IP55 dust-and-water ingress, tested in a NABL-accredited lab · founder trained as ISO 13485 internal auditor

NEXT 12 MO

ISO 13485 quality system + CDSCO manufacturing licence (Form MD-5) · WHO PQS E003 submission



Real payload – a blood bag in the 2 L chamber

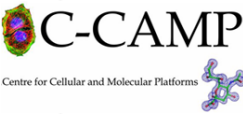
Who buys it. Who backs it.

Beckman Coulter, Roche India and Solumatix are the three commercial anchors (pages 5 and 8). Around them, clinics, hospitals, dairy and research — with units reporting from five Indian states.

CUSTOMERS, PILOTS & DEPLOYMENTS

 Diagnostics • bought after six months	 Cold-chain-as-a-service • Kolkata • starting	 SELCO Foundation Primary-health outreach	 Swami Vivekananda Youth Movement District hospital	 Dairy & veterinary
 Eye care	 Rural health, Rajasthan	 TIGS Tata Institute for Genetics and Society Genetics research	 Screening	 30-unit order • advance received

INCUBATED & FUNDED BY

 SEED fund	 TIDE 2.0	 NIDHI PRAYAS	 Incubation	 Incubation
 Grant	 IQ 2.0 grant	 Leaders in Innovation Fellowship	 DPIIT recognised	 Karnataka • Tamil Nadu • Maharashtra • Gujarat • Uttar Pradesh

A market exists, so the device **does not live on grants.**

\$1.2 billion addressable; \$0.7 billion in the regions our distributors reach. Commercial buyers — labs, pharma, blood banks — carry the company; grant funding buys the certification and evidence that let public programmes buy at \$500 a unit.

\$1.2^B

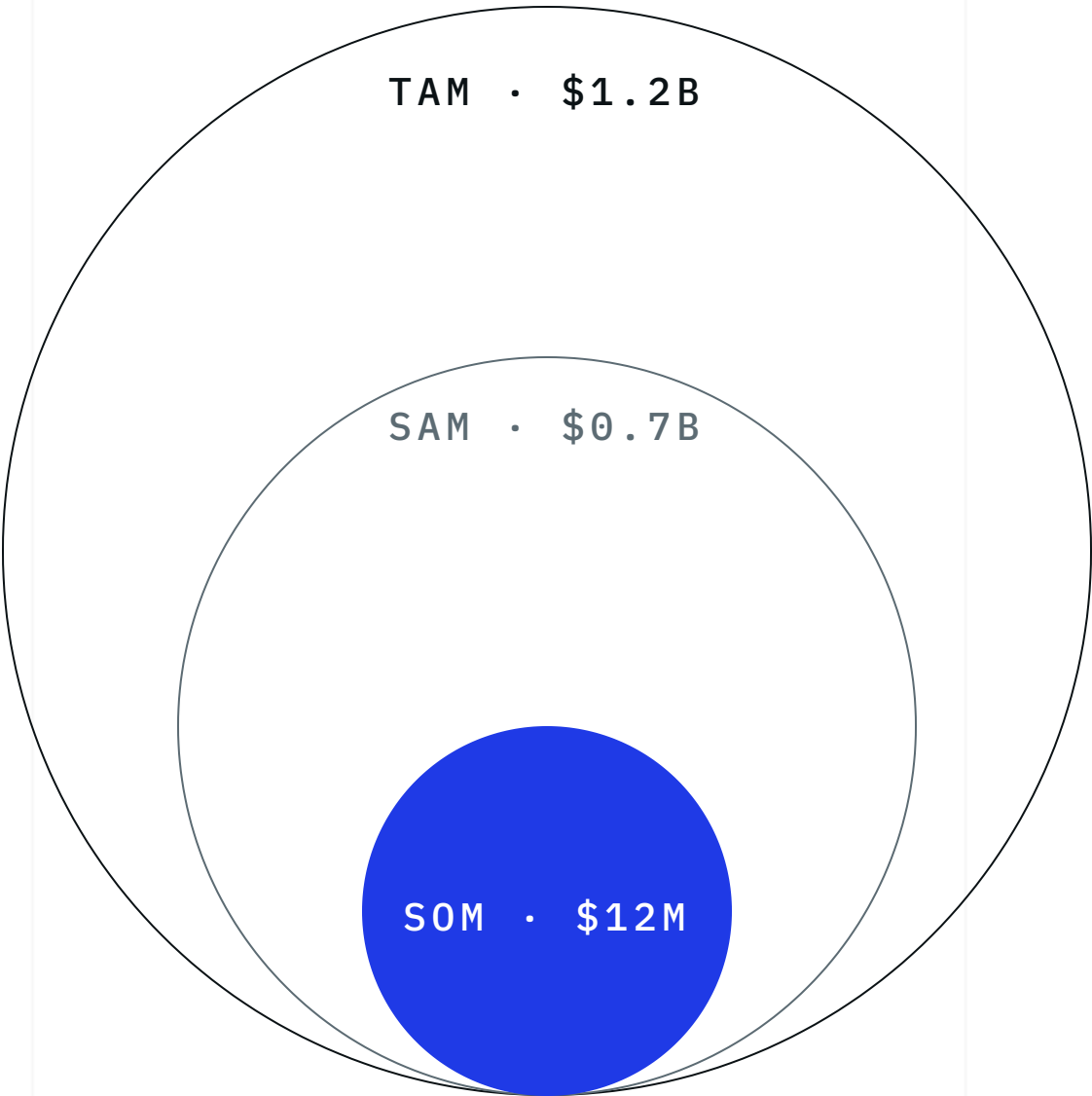
TAM — about 1 million last-mile cold-chain devices in use worldwide × ~\$1,200 average lifetime value at market prices. Company bottom-up estimate.

\$0.7^B

SAM — India, the Middle East, Central Asia, Africa and Southeast Asia: the geographies our distributor model can reach.

\$12^M

SOM — 1% of TAM a year by year five, deliberately modest. Our year-five plan of \$7.9M sits inside it.



Context: pharma spends \$21.3B a year on cold-chain logistics (IQVIA, 2025). In Kazakhstan and Uzbekistan about 9 in 10 medicines are imported, so cold-chain products travel a long way to the patient — which is why Central Asia sits inside the SAM. TAM is our separate bottom-up estimate of the last-mile device market.

Made anywhere. Serviced by local technicians.

It is solid-state — no compressor line, no refrigerant to handle — so it can be built in India today and assembled in any region when volume or local rules call for it. Local technicians are trained with every deployment, and 4G diagnostics tell them what to fix before they travel. The need is wherever the grid is not (top row); on our model, this is what each unit adds (bottom row).

3.2^B

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WHO / UNICEF, July 2026

1^{in 2}

hospitals in sub-Saharan Africa lack reliable electricity; 15% of facilities have none

WHO • World Bank • IRENA, 2023

1,900

doses protected per unit, per year – 7,500 doses handled on outreach × 25% baseline wastage

Company model

43^M

doses a year by FY30 across 22,800 units

Company plan, August 2026

10,200^t

CO₂e avoided a year by FY30 – no compressor, no refrigerant; about 0.45 t per device. Cooling already drives 7% of global emissions.

Company model • UNEP, 2023

From a 50 ml prototype to a deployed platform.

Five years, each version rebuilt from field data — from a bench prototype to ten units with SELCO Foundation to a commercial device.
Next: new countries, starting with the Gulf.



2021
Genesis
50 ml solar-powered unit. Proof of concept.



2023
Field trials
10 units deployed with SELCO Foundation.



2024
Version 3.0
Redesigned from field data. CDSCO test licence.



2025
Commercial-ready
Analytics on board. Tested at TÜV SÜD. Patent filed.



2026 →
Scale
2.5 L, 5 L and 10 L. International expansion.

Today the device. Tomorrow the chain.

One cold chain, end to end. Samples reach the lab in a 2 L; doses leave the manufacturer in a 10 L, cross the country in a cold container, then step down — a 5 L on a cycle, a 2 L on a health worker's back — to a child under a tree, with one temperature record the whole way.

THE LONG VIEW

One provider and one temperature record from factory cold room to vaccination session: transport, last-mile devices and site storage, all monitored, all auditable. In this plan: the 2 L today, the 5 L in FY27, the 10 L in FY28. The container and site storage are the ten-to-twenty-year view. The biologics decade needs that chain of custody, and the record that proves it.



Design, electronics and firmware. All developed in-house.



Suraj J R

CO-FOUNDER

Built the cold engine and next-gen controller

Shruti

FIRMWARE

Embedded and edge-AI control

Ankita Mittal

CEO

ISB • market entry, first institutional deployments

Prerit Mittal

FOUNDER

IIM-A • Royal Academy of Engineering Fellow

Chairman • Harkesh Kumar Mittal – architect of India's national incubation programme (NIDHI); chaired DPIIT's Start-up India seed fund committee

Group portrait composed from the team's individual photographs

\$350k over eighteen months.

A programme grant that buys certification, evidence and a hundred units in public health systems. It sits beside a separate \$1.1M equity round that funds manufacturing.

30%	100 units placed with two public health systems, local technicians trained with every deployment
30%	Comparative field study — 30 units against 30 ice-pack carriers, ethics-approved, data published
25%	WHO PQS E003 testing and submission
15%	Programme management, telemetry and open data

BEYOND CAPITAL

Introductions to national immunisation programmes and UN procurement, and a seat where prequalification and procurement standards are written. Every unit reports its readings live, so the funder sees the evidence as it is generated.

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The platform the grant de-risks – 5 L for primary health centres, FY27